



Data Analytics

Turning raw data into actionable business insight

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Data Analytics examines datasets to draw conclusions and identify trends, helping organizations make informed, evidence-based decisions.

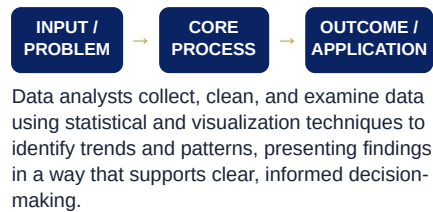
● DEFINITION

The process of examining data sets to draw conclusions about the information they contain, often using specialized systems and software.

● WHY IT MATTERS

Organizations generate enormous volumes of data — analytics converts that raw data into actionable insights that drive better business and operational decisions.

● WORKING PRINCIPLE



● QUICK FACTS

- ▶ Data analytics helped major retailers famously predict customer purchasing patterns before customers themselves were aware of them
- ▶ Business analytics dashboards can now update in real time, rather than the periodic reports of the past

● KEY TERMS

Descriptive analytics Predictive analytics
KPI Data visualization

● CORE CONCEPTS

- ▶ Descriptive, predictive, prescriptive analytics
- ▶ Data visualization
- ▶ Key performance indicators
- ▶ Data-driven decision-making

● APPLICATIONS

- ▶ Business performance tracking
- ▶ Customer behavior analysis
- ▶ Operational efficiency analysis
- ▶ Marketing campaign optimization

● ADVANTAGES

- ▶ Enables evidence-based business decisions
- ▶ Identifies trends and opportunities not obvious from raw data
- ▶ Applicable across virtually every business function

CAREER OPPORTUNITIES

Data analyst, Business intelligence analyst, Marketing analyst, Operations analyst

RESEARCH AREAS

Real-time analytics systems, Automated insight generation, Predictive analytics modeling

● REAL-LIFE EXAMPLES

- ▶ Retail sales trend dashboards
- ▶ Website traffic analysis
- ▶ Customer churn prediction

● LIMITATIONS

- ▶ Poor quality or incomplete data undermines analysis
- ▶ Requires clear communication to translate insight into action
- ▶ Can be misused to support predetermined conclusions

INDUSTRY APPLICATIONS

Retail, Marketing, Finance, Healthcare

FUTURE SCOPE

AI-powered analytics tools are making sophisticated data analysis faster and more accessible to non-technical business users.

● CURRENT TRENDS

Growing use of AI-powered analytics platforms that automatically surface insights without requiring manual analysis.

● SUMMARY

Data analytics examines data to uncover trends and insights, helping organizations make informed, evidence-based decisions across every function.

